

Degly Sebastián Pava Pava

✉ deglypavapava@gmail.com 🌐 <https://deglypava.com> 🎓 IMLEX and Systems and Computer Engineer

From: August 2020
To: June 2023
(Currently in Japan)

Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- *Quality Board Member and Representative*
- *Toyohashi University of Technology (Japan), University of Eastern Finland (Finland), University Jean Monnet Saint-Etienne (France) and KU Leuven (Belgium)*

Photonics | Physical Optics | Realtime 3DRendering | Computational Imaging | CUDA | Computer Graphics | Machine Learning | ARKit | ARCore | VR/AR.

WORK EXPERIENCE

Company Owner
(CEO – CTO)

Omni Applications L.L.C. (Florida, U.S.A) – Lucid Dreams Ltd. (Bogotá, Colombia)

From: January 2018
To: August 2020

Lead XR Developer | Hackspace for Innovation and Visualization in Education

Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)

- Part of THE HIVE Research Laboratory working in The HoloBrain and MR Manikin / OCS projects. In charge of the creation of multiple 3D Experiences and their integration into a Mixed Reality environment while using concepts of 3D visual computing, computational geometry, VR, ARKit / ARCore and MR toolkits within the Unity 3D Engine.
- **Tech: Unity | Microsoft HoloLens 1&2 | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy | Azure**

From: February 2017
To: September 2019

Lead Developer and Engineer at Pontifical Xavierian University

Department of Industrial Engineering and Systems Engineering at Pontifical Xavierian University, Bogota (Colombia)

- By using the NetLogo programming language I developed a simulation of a production planning, scheduling, and inventory control system used to manage manufacturing processes, also known as a Material Requirements Planning. It is currently being used in production classes to teach new students about MRP and scheduling.

Tech: NetLogo | Anylogic | Microsoft Office Suite | Adobe Suite | Research Techniques

EDUCATION AND TRAINING

Master's studies:

IMLEX + MANUTECH Slight Scholarship + ERASMUS Mundus Japan Scholarship

Bachelor studies:

Systems and Computer Engineer. Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia.

Trained in multiple research groups:

HIVE Research Lab – ELAP (Emerging Leaders of the Americas Program) Scholarship

The University of British Columbia, Vancouver, B.C. (Canada)

From: January 2019
To: April 2019

- Emerging Leaders of the Americas Program Scholarship winner offered by the Canadian Government.
- Visiting International Research Scholar Fellowship.
- Developed neuroanatomy-based applications while making use of MR and VR technologies for research and educational purposes.

From: June 2018
To: December 2018

Gold IronHacks Purdue-UNAL 2018 Fellowship – RCODI

Purdue University, West Lafayette, Indiana (United States)

- Participated on the team that organized the Gold IronHacks Purdue-UNAL Hackathon 2018.
- Created and managed parts of the research such as surveys and privacy agreements.
- Part of the judges that would choose the best implementation of Open Data within a participant created website. Using a defined set of attributes, we analysed how the data was implemented by students.

Digital skills	Extended Realities	Programming	Optics / Physics	Research and Tools
	Unity Blender ThreeJS	C# Python C++	Photonics Colorimetry	ImageJ3D Medical Software
	AWS Azure MatlabCloud	JavaScript NetLogo	Eye Tracking Robotics	TensorFlow TensorBoard
	Vuforia MRTK Arduino	Git (GitHub) SCRUM	Spectral Imaging	GIMP2 Adobe Suite

PERSONAL SKILLS

Spanish

Mother Tongue

French

Basic Writing and Speaking Skills

English

IELTS Academic (8.0) and iBT TOELF (103 points)

Japanese

Furigana / Basic Speaking

AWARDS AND RESEARCH PROJECTS

- **HoloBrain 2.0 – ELAP Scholarship - THE HIVE at The University of British Columbia**

A mixed reality brain simulation to allow educators to teach the anatomy of the brain with a holographic model, using 3D reconstructions of basal ganglia nuclei which were obtained from MRI scans. Developed a volumetric rendering that could behave as a 3D Mesh and be interactable with hand gestures so the user could cut the brain in real time to have a better idea of its parts and how it was internally organized. A paper is being written on how 3D Visualizations are better than 2D materials.

- **MR. Manikin / Online Clinical Simulation – VIRS Research Fellowship - THE HIVE at The University of British Columbia**

A mobile AR application that would enable medical students to interact with a critical care patient. Developed a 3D model of the patient, used Vuforia Engine to place it on the real world and Photon Networking within Unity 3D to send instructions from the Teacher Facilitator (iPad or PC) to the student interacting with the patient (Android or iOS). The application is currently in beta, soon to be published on the App Stores and its being used on multiple medical research projects.

- **Energy Modelling Tool – RCODI's Research Fellowship - Purdue's Research Center for Open Digital Innovation**

HTML and AWS research-based software focused on understanding what civil engineers and architects need to find the best way to build an energy efficient building by using data obtained from EnergyPlus processing, Open Data and MATLAB Cloud Computing. All of these tools were used for cost calculation and component selection.

- **Winner of Purdue-UNAL Gold IronHacks 2017**

Developed a website with a mashup that utilized Open Data to help students rent houses near the University of Illinois, Chicago. Open Data was used to create interactive visualizations to give insights and support optimal decisions. The project was developed through Git and judged by a team of experts from Purdue University. Later on I had the chance to join RCODI and help with the research project that analysed how the students who participated, used the Open Data provided in addition to visualizations such as D3.js.

- **Lead developer, Simulation of TransMilenio – Full Scholarship at Universidad Nacional de Colombia**

Developed a NetLogo simulation of Bogota's transport service TransMilenio to try to optimize how buses should be deployed to improve the quality of service. The software simulated how the system works, showing its flaws and delays. I also propose an algorithm to improve its quality. The project required multiple measurement visits to the system and optimizations to the model. In the end we were able to propose a different approach to how the buses are deployed in order to improve the bad service provided.